MATH 111 Calculus 1

# The chain rule

## and implicit differentiation

Week 4 · Lecture 10 · §2.5, §2.6

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

### While you wait

Sit with your team. A weather balloon rises at 5 metres per second. Higher up the air is colder: about 6.5 °C colder for every kilometre of height. How fast is the temperature *at the balloon* falling — in degrees per minute?

STAND UP

"The derivative of $\sin(x^3)$ is $\cos(x^3)$."

**Stand up if you think this is TRUE.**

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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## Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

### Verdict

**FALSE.** Check at $x = 0$: the true slope there is $\lim_{h\to 0} \dfrac{\sin(h^3) - 0}{h} = \lim_{h\to 0} \dfrac{\sin(h^3)}{h^3} \cdot h^2 = 1 \cdot 0 = 0$, but $\cos(0^3) = 1$. Something is missing: the derivative of the *inside*, $3x^2$. Today's rule says exactly what is missing, and why.

# Rates multiply along a chain — the balloon

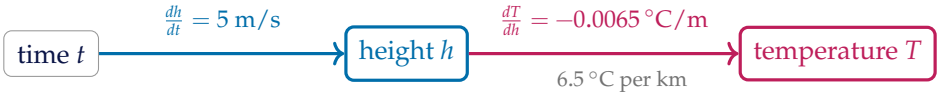
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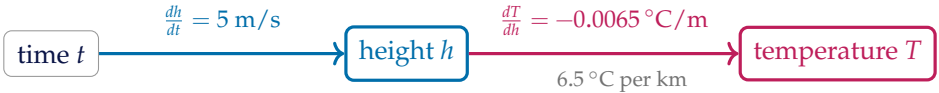
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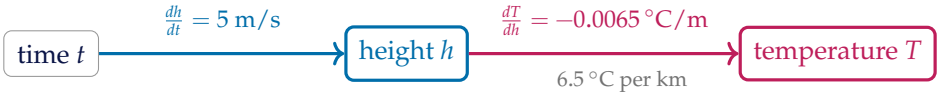



**Let the units think for you:** $\dfrac{dT}{dt} = \dfrac{dT}{dh} \cdot \dfrac{dh}{dt}$, $\qquad \dfrac{^\circ\text{C}}{\text{s}} = \dfrac{^\circ\text{C}}{\text{m}} \cdot \dfrac{\text{m}}{\text{s}}$.

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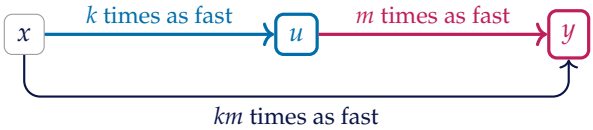



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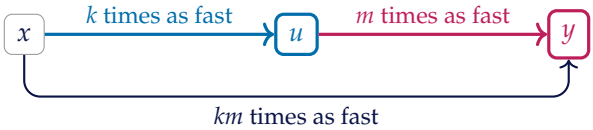
**Here:** $\dfrac{dT}{dt} = (-0.0065) \cdot 5 = -0.0325\,^\circ\text{C/s} = -1.95\,^\circ\text{C/min}.$

The temperature at the balloon falls by about 2 °C a minute.

# Rates multiply along a chain — in general

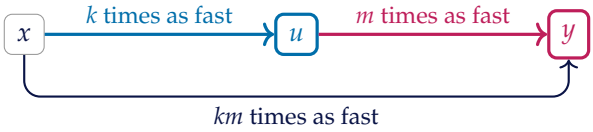


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If $u$ changes $k$ times as fast as $x$, and $y$ changes $m$ times as fast as $u$, then $y$ changes $km$ times as fast as $x$: rates multiply along a chain.

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**Leibniz's picture**

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$ — as if the $du$ cancelled.

It does not cancel: $du$ is not a number (Lecture 8). But the formula the picture suggests is *true*, and that is the chain rule.

**The chain rule**

If $g$ is differentiable at $x$ and $f$ is differentiable at $g(x)$, then $F(x) = f\big(g(x)\big)$ is differentiable at $x$, and

$$F'(x) = f'\big(g(x)\big) \cdot g'(x).$$

In Leibniz's notation, with $y = f(u)$ and $u = g(x)$: $\dfrac{dy}{dx} = \dfrac{dy}{du} \cdot \dfrac{du}{dx}$ ($dy/du$ taken at $u = g(x)$).

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**How to read it: outside, then inside.** Differentiate the *outside* function, leaving the inside alone; then multiply by the derivative of the *inside*:

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**How to read it: outside, then inside.** Differentiate the *outside* function, leaving the inside alone; then multiply by the derivative of the *inside*:

$$\frac{d}{dx} f\big(g(x)\big) = \underbrace{f'}_{\substack{\text{derivative} \\ \text{of outside function}}} \underbrace{\big(g(x)\big)}_{\substack{\text{evaluated at} \\ \text{inside function}}} \cdot \underbrace{g'(x)}_{\substack{\text{derivative} \\ \text{of inside function}}}$$

# Three examples (1) — a power of a polynomial

**(a)** $y = (x^2 + 5x)^8$, that is $y = \underbrace{u^8}_{\substack{\text{outside} \\ \text{function}}}$ with $u = \underbrace{x^2 + 5x}_{\substack{\text{inside} \\ \text{function}}}$.

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**The power rule with a chain:** $\dfrac{d}{dx}\big[g(x)\big]^n = n\big[g(x)\big]^{n-1} \cdot g'(x)$ for every real $n$ (where the right side is defined).

# Three examples (2) — a root

**(b)** $y = \sqrt{x^2+9} = (x^2+9)^{1/2}$, that is $y = \underbrace{u^{1/2}}_{\text{outside function}}$ with $u = \underbrace{x^2+9}_{\text{inside function}}$.

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Rewrite the root as a power first (Lecture 9): the power rule with a chain does the rest.

# Three examples (3) — which one is outside?

**(c) Back to the warm-up**, and the same two functions in the other order:

$y = \sin(x^3)$:

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Same two functions, composed in the other order: different functions, different derivatives. The first question is always "which one is outside?"

# Question 21

Let $h(x) = f(g(x))$, and suppose $g(1) = 3$, $g'(1) = 4$, $f'(1) = 7$, $f'(3) = 5$. What is $h'(1)$?

**A.** 28

**B.** 5

**C.** Not enough information: we would need $g'(3)$.

**D.** 20

## In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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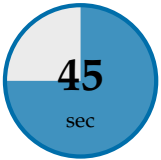
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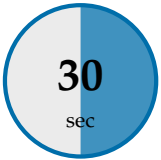
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**Hands down. Thank you.**



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## Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

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## Answer

**D.** $h'(1) = f'\big(g(1)\big) \cdot g'(1) = f'(3) \cdot g'(1) = 5 \cdot 4 = 20$. A is $f'(1)\,g'(1)$: the outside derivative fed the wrong number — $f'$ wants $g(1) = 3$, not 1. B is $f'(3)$ alone: the inside forgotten, the warm-up's mistake in numbers. C wants $g'$ at $g(1) = 3$ — the roles reversed. In fact the *inside* derivative is taken at $x = 1$ and the *outside* at $g(1)$; nothing is missing.

# A chain with three links

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$$y' = \underbrace{\cos}_{\substack{\text{derivative}\\ \text{of outside}}} \underbrace{(\cos 2x)}_{\substack{\text{evaluated}\\ \text{at the rest}}} \cdot \Big( \underbrace{-\sin}_{\substack{\text{derivative}\\ \text{of middle}}} \underbrace{2x}_{\substack{\text{evaluated}\\ \text{at inside}}} \Big) \cdot \underbrace{2}_{\substack{\text{derivative}\\ \text{of inside}}} = -2 \sin 2x \, \cos(\cos 2x).$$

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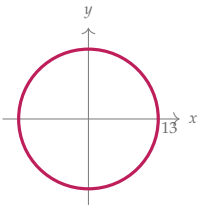
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**Your turn, in the tutorial**

$\dfrac{d}{dx}\sqrt{1 + \sin 3x}$ and $\dfrac{d}{dx}\cos^2(5x)$ — find the *outside* before you write anything.

# Functions defined implicitly

So far, functions were given *explicitly*: $y = f(x)$. Some are defined implicitly, by a relation between $x$ and $y$ — such as the circle $x^2 + y^2 = 169$.

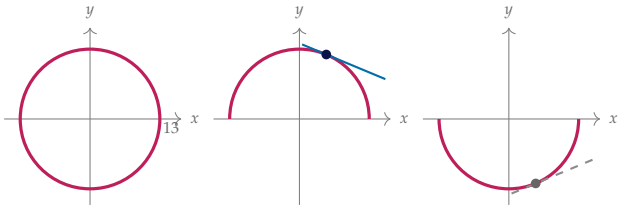



(a) $x^2 + y^2 = 169$

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Solving for $y$: $y = \pm\sqrt{169 - x^2}$. The equation determines two functions, $f(x) = \sqrt{169 - x^2}$ and $g(x) = -\sqrt{169 - x^2}$: the upper and the lower half of the circle.



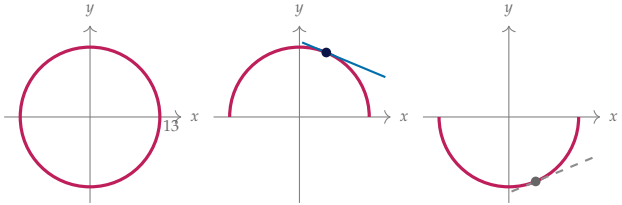

(a) $x^2 + y^2 = 169$ (b) $y = f(x)$ (c) $y = g(x)$

tangents at $(5, 12)$ and $(5, -12)$: frame 14

# Functions defined implicitly

So far, functions were given *explicitly*: $y = f(x)$. Some are defined implicitly, by a relation between $x$ and $y$ — such as the circle $x^2 + y^2 = 169$.

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**Implicit differentiation**

We never need $f$ or $g$.

**1.** Differentiate both sides with respect to $x$, treating $y$ as a function of $x$: each $y$ you differentiate brings a $\frac{dy}{dx}$ (chain rule) — $\frac{d}{dx}(y^2) = 2y\frac{dy}{dx}$.

**2.** Solve for $\frac{dy}{dx}$.

# The circle, worked

**Step 1.** $\dfrac{d}{dx}\left(x^2 + y^2\right) = \dfrac{d}{dx}(169)$, that is $2x + 2y\dfrac{dy}{dx} = 0$ ($y^2$ is "something squared"; the something depends on $x$).

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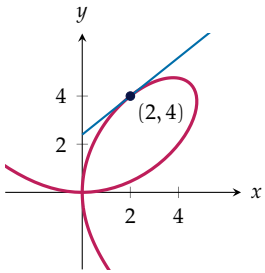
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School geometry agrees: the radius to $(5,12)$ has slope $\frac{12}{5}$, the tangent $-\frac{5}{12}$, and $\frac{12}{5}\cdot\left(-\frac{5}{12}\right) = -1$: perpendicular.

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$x^3 + y^3 = 9xy$ (a *folium of Descartes*): not solvable for $y$ in any useful way. Differentiate it as it stands.

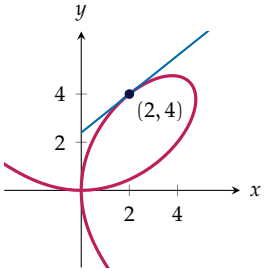



$x^3 + y^3 = 9xy$ and its tangent at $(2, 4)$

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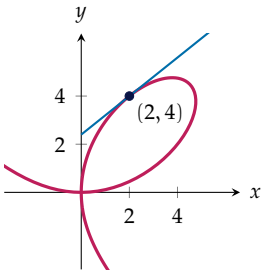

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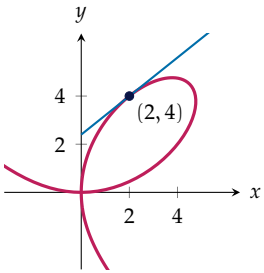
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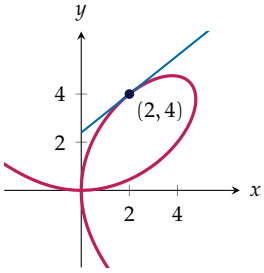
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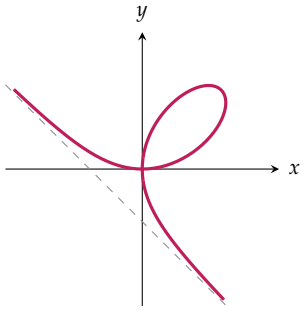
Where is the tangent horizontal, away from the origin? $y' = 0$ means $3y = x^2$ — the tutorial.




$x^3 + y^3 = 9xy$ and its tangent at $(2,4)$

**A little history · 1638**

**1638. Descartes** studied the curve $x^3 + y^3 = 3axy$ — ours, with $3a = 9$. He found its shape in the first quadrant correctly, but believed that the leaf was repeated in every quadrant, like the four petals of a flower.

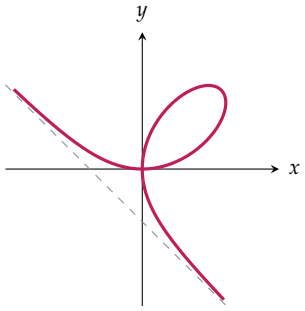



$x^3 + y^3 = 3axy$ with $a = 1$: one leaf, and the asymptote $x + y + a = 0$ (dashed)

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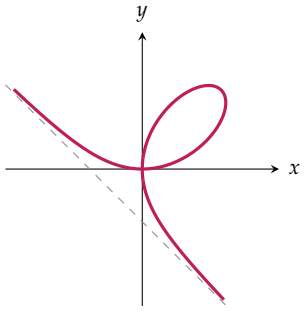

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Its tangent was a challenge of the day — put to Fermat in one account, to Roberval in another. In the first, Fermat solved it easily, which Descartes could not. With implicit differentiation it took us two lines.

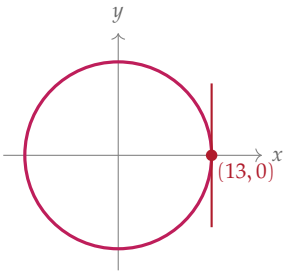



$x^3 + y^3 = 3axy$ with $a = 1$: one leaf, and the asymptote $x + y + a = 0$ (dashed)

**The assumption behind the method**

Implicit differentiation assumes that, near the point, the equation determines $y$ as a differentiable function of $x$. We assume it wherever we use the method — and it can fail: on the circle $x^2 + y^2 = 169$ at $(13, 0)$ the tangent is vertical, step 1 reads $26 + 0 \cdot y' = 0$, no value of $y'$ works, and near $(13, 0)$ the circle is not the graph of *any* function of $x$.

The theorem that says *when* the assumption holds — the *implicit function theorem* (usually credited to Cauchy; for real variables, Ulisse Dini, Pisa, 1877–78) — belongs to a later course.




$x^2 + y^2 = 169$: a vertical tangent at $(13, 0)$

**A little history · 1676–1823**

**Leibniz**, in a memoir of 1676, differentiates $\sqrt{a + bz + cz^2}$ by first putting $x = a + bz + cz^2$ — an "inside" — and then differentiating the root of $x$: as far as historians can tell, the first use of the chain rule (with mistakes in the calculation). The notation that makes the rule look obvious, $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$, is his.

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The first modern statement is usually credited to **Lagrange**, in the *Théorie des fonctions analytiques* of 1797 — the book of the notation $f'(x)$ (Lecture 7) — and the rule is in **Cauchy**'s *Résumé des leçons* of 1823, the book that gave the derivative its limit.

## Today in two lines

1. Rates multiply along a chain: $\big[f(g(x))\big]' = f'(g(x)) \cdot g'(x)$, $\dfrac{dy}{dx} = \dfrac{dy}{du}\dfrac{du}{dx}$.
   Outside first, inside left alone, then multiply by the inside's derivative.

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**This week:**

Homework: §2.5 and §2.6 — read and work out the examples. Next: the revision lecture — then inverse functions and their derivatives (§6.1).

# EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *"The derivative of* $\sin(x^3)$ *is not* $\cos(x^3)$ *because . . . " (one sentence, or the missing factor).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.